Mock central exam: solutions and point scheme

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The point scheme follows the style of the official *correctievoorschrift*: each bullet earns the number of points shown at the right. A different correct method earns full points. All answers were checked with SymPy/Python.

General rules (adapted from the CvTE marking rules for mathematics):

- An answer without the required working (calculation, explanation or reasoning) earns **0 points** for that question, unless the scheme says otherwise.
- For an *exact* question, using a decimal approximation in an essential step costs at least the points for that step.
- Rounding too early or to the wrong precision: **1 point** deduction, at most once per question.
- With a graphing-calculator step, write down what you entered (function, equation, window). If missing: **1 point** deduction.
- A consequential error (*doorwerkende fout*) is penalised once; later steps that correctly use the wrong value can still earn points.

Grade. The CE grade is computed from the score S (out of 75) as grade $= 9 \cdot \frac{S}{75} + N$, where the *N-term* is set by the CvTE after each exam (between 0.0 and 2.0; $N = 1$ is neutral), with extra rules near the bottom and top of the scale. For practice use $N = 1$: a 5.5 then needs about 38 points and a 9 about 67.

Part 1

Question 1 (4p)

$f'(x) = e^{-\frac{1}{2}x} + x \cdot (-\frac{1}{2})e^{-\frac{1}{2}x} = e^{-\frac{1}{2}x}\left(1 - \frac{1}{2}x\right)$. $f'(x) = 0 \iff 1 - \frac{1}{2}x = 0$ (since $e^{-\frac{1}{2}x} > 0$) $\iff x = 2$. f' changes from positive to negative at $x = 2$, so this is a maximum: top $(2, \frac{2}{e})$.

- Product rule and chain rule correctly applied: **1**
- $f'(x) = e^{-\frac{1}{2}x}(1 - \frac{1}{2}x)$ or equivalent: **1**
- $x = 2$, with the reason that $e^{-\frac{1}{2}x} \neq 0$: **1**
- $f(2) = 2e^{-1}$, coordinates $(2, \frac{2}{e})$: **1**

Question 2 (4p)

Integration by parts, or check the antiderivative $F(x) = -2xe^{-\frac{1}{2}x} - 4e^{-\frac{1}{2}x}$: $F'(x) = -2e^{-\frac{1}{2}x} + xe^{-\frac{1}{2}x} + 2e^{-\frac{1}{2}x} = xe^{-\frac{1}{2}x}$. ✓ Since $f \geq 0$ on $[0, 2]$:

$$\text{area} = F(2) - F(0) = \left(-4e^{-1} - 4e^{-1}\right) - (-4) = 4 - \frac{8}{e}.$$

- Correct antiderivative (by parts, or guessed and checked by differentiating): **2**
- Setting up $\int_0^2 f(x)\,dx = F(2) - F(0)$: **1**
- Answer $4 - \frac{8}{e}$ (≈ 1.057): **1**

Question 3 (5p)

$f''(x) = -\frac{1}{2}e^{-\frac{1}{2}x}(1 - \frac{1}{2}x) + e^{-\frac{1}{2}x} \cdot (-\frac{1}{2}) = e^{-\frac{1}{2}x}\left(\frac{1}{4}x - 1\right)$. $f''(x) = 0 \iff x = 4$, and f'' changes sign there: inflection point $(4, 4e^{-2})$. Slope: $f'(4) = e^{-2}(1 - 2) = -e^{-2}$. Tangent:

$y = 4e^{-2} - e^{-2}(x-4) = e^{-2}(8-x)$. For $y = 0$: $x = 8$. ✓

- $f''(x)$ correct: **2**
- $x = 4$ and $f(4) = 4e^{-2}$: **1**
- $f'(4) = -e^{-2}$: **1**
- Equation of the tangent and $y = 0 \Rightarrow x = 8$: **1**

Part 2

Question 4 (5p)

$g'(x) = 2\cos 2x - 2\sin x = 2(1 - 2\sin^2 x) - 2\sin x = -2(2\sin^2 x + \sin x - 1) = -2(2\sin x - 1)(\sin x + 1)$. $g'(x) = 0 \iff \sin x = \frac{1}{2}$ or $\sin x = -1 \iff x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$. Values: $g(\frac{\pi}{6}) = \frac{1}{2}\sqrt{3} + \sqrt{3} = \frac{3}{2}\sqrt{3}$; $g(\frac{5\pi}{6}) = -\frac{3}{2}\sqrt{3}$; $g(\frac{3\pi}{2}) = 0$; endpoints $g(0) = g(2\pi) = 2$. Since $\frac{3}{2}\sqrt{3} \approx 2.60 > 2$, the maximum is $\frac{3}{2}\sqrt{3}$. (At $x = \frac{3\pi}{2}$, g' does not change sign: a horizontal inflection point, not an extremum.)

- $g'(x) = 2\cos 2x - 2\sin x$: **1**
- Using $\cos 2x = 1 - 2\sin^2 x$ to get a quadratic in $\sin x$: **1**
- $\sin x = \frac{1}{2}$ or $\sin x = -1$, with the x-values: **1**
- $g(\frac{\pi}{6}) = \frac{3}{2}\sqrt{3}$ computed exactly: **1**
- Comparing with the other candidates, including the endpoints: **1**

Question 5 (3p)

$g(x) = 2\sin x\cos x + 2\cos x = 2\cos x(\sin x + 1) = 0 \iff \cos x = 0$ or $\sin x = -1$. $\cos x = 0$: $x = \frac{\pi}{2}$ or $x = \frac{3\pi}{2}$. $\sin x = -1$: $x = \frac{3\pi}{2}$. Zeros: $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$.

- Factorising $2\cos x(\sin x + 1)$: **1**
- Both cases solved: **1**
- Answer $\frac{\pi}{2}, \frac{3\pi}{2}$ (and nothing else): **1**

Question 6 (4p)

$$I = \pi \int_0^{\pi} 4\sin^2 x\, dx = \pi \int_0^{\pi} (2 - 2\cos 2x)\, dx = \pi\,[2x - \sin 2x]_0^{\pi} = 2\pi^2.$$

- Setting up $\pi \int_0^{\pi} (2\sin x)^2\, dx$: **1**
- Rewriting $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$: **1**
- Antiderivative $2x - \sin 2x$: **1**
- Answer $2\pi^2$: **1**

Part 3

Question 7 (4p)

c: $(x-3)^2 + (y-2)^2 = 5$, centre $M(3, 2)$, radius $\sqrt{5}$. ℓ_k: $2x - y + k = 0$. Tangent iff $d(M, \ell_k) = \sqrt{5}$:

$$\frac{|6 - 2 + k|}{\sqrt{5}} = \sqrt{5} \iff |4 + k| = 5 \iff k = 1 \text{ or } k = -9.$$

(Alternative: substitute $y = 2x + k$, get $5x^2 + (4k - 14)x + (k^2 - 4k + 8) = 0$, and set the discriminant $-4(k^2 + 8k - 9) = 0$.)

- Centre and radius: **1**
- Distance formula (or substitution) set up: **1**
- $|4 + k| = 5$ (or the discriminant equation): **1**
- $k = 1$ or $k = -9$: **1**

Question 8 (4p)

$\overrightarrow{MP} = (-2, -1)$, $\overrightarrow{MQ} = (1, 2)$. $\cos\angle PMQ = \dfrac{-2-2}{\sqrt{5}\cdot\sqrt{5}} = -\dfrac{4}{5}$, so $\angle PMQ \approx 143.13°$. The tangents are perpendicular to the radii, so in quadrilateral $MPTQ$: $\angle MPT = \angle MQT = 90°$ and $\angle PTQ = 360° - 90° - 90° - 143.13° \approx 36.87°$. Answer: $37°$.

- Using the radius vectors (or the tangent directions $(1, -2)$ and $(2, -1)$): **1**
- Inner product and cos $= -\frac{4}{5}$ (or $\frac{4}{5}$ for the tangent directions): **1**
- Argument that the angle at T is $180°$ minus the angle at M (or correctly choosing the acute angle between the tangents): **1**
- $37°$: **1**

Part 4

Question 9 (4p)

Horizontal tangent: $y'(t) = 0$ and $x'(t) \neq 0$. $y'(t) = 2\cos 2t = 0 \iff 2t = \frac{\pi}{2} + k\pi \iff t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$. $x'(t) = -\sin t \neq 0$ at these values. ✓ Points: $t = \frac{\pi}{4}$: $(\frac{1}{2}\sqrt{2}, 1)$; $t = \frac{3\pi}{4}$: $(-\frac{1}{2}\sqrt{2}, -1)$; $t = \frac{5\pi}{4}$: $(-\frac{1}{2}\sqrt{2}, 1)$; $t = \frac{7\pi}{4}$: $(\frac{1}{2}\sqrt{2}, -1)$.

- Condition $y'(t) = 0$ (and checking $x'(t) \neq 0$): **1**
- $y'(t) = 2\cos 2t$: **1**
- Four values of t: **1**
- Four points exactly: **1**

Question 10 (4p)

$(x, y) = (0, 0)$: $\cos t = 0 \Rightarrow t = \frac{\pi}{2}$ or $t = \frac{3\pi}{2}$ (and then $\sin 2t = 0$ ✓). Velocity vector $(x'(t), y'(t)) = (-\sin t, 2\cos 2t)$: at $t = \frac{\pi}{2}$: $(-1, -2)$; at $t = \frac{3\pi}{2}$: $(1, -2)$. $\cos\varphi = \dfrac{(-1)(1) + (-2)(-2)}{\sqrt{5}\sqrt{5}} = \dfrac{3}{5}$, $\varphi \approx 53.13°$. Answer: $53°$.

- Both values of t: **1**
- Both velocity vectors: **1**
- Inner product formula: **1**
- $53°$: **1**

Question 11 (3p)

$y = \sin 2t = 2\sin t\cos t$, so $y^2 = 4\sin^2 t\cos^2 t = 4\cos^2 t(1 - \cos^2 t) = 4x^2(1 - x^2)$.

- Using $\sin 2t = 2\sin t\cos t$: **1**
- Using $\sin^2 t = 1 - \cos^2 t$: **1**
- Substituting $x = \cos t$ to reach the required form: **1**

Part 5

Question 12 (3p)

$V'(t) = 30 \cdot 0.1e^{-0.1t} = 3e^{-0.1t}$. $3 - 0.1V = 3 - 3(1 - e^{-0.1t}) = 3e^{-0.1t}$. Equal for all t. ✓
$V(0) = 30(1-1) = 0$. ✓

- $V'(t)$ correct: **1**
- Right-hand side simplified and compared: **1**
- Initial condition checked: **1**

Question 13 (4p)

$V_{n+1} = V_n + 2(3 - 0.1V_n)$. $t = 0$: $V_0 = 0$. $t = 2$: $V_1 = 0 + 2 \cdot 3 = 6$. $t = 4$: $V_2 = 6 + 2(3 - 0.6) = 10.8$.
Exact: $V(4) = 30(1 - e^{-0.4}) \approx 9.890$. Difference: $10.8 - 9.890 \approx 0.91$ litres.

- Euler recurrence correctly set up: **1**
- $V_1 = 6$: **1**
- $V_2 = 10.8$: **1**
- Difference 0.91: **1**

Part 6

Question 14 (3p)

$P(D < 9.90) = P(D > 10.10)$ by symmetry; normalcdf$(-10^{99}, 9.90, 10, 0.05) \approx 0.02275$.
$P(\text{rejected}) = 2 \cdot 0.02275 \approx 0.046$.

- Recognising both tails (or $1 - P(9.90 < D < 10.10)$): **1**
- Correct calculator command with parameters written down: **1**
- 0.046: **1**

Question 15 (4p)

X = number of rejected bolts in the box, binomial with $n = 50$, $p = 0.0455$ (unrounded from question 14). $P(X \geq 3) = 1 - P(X \leq 2) = 1 - \text{binomcdf}(50, 0.0455, 2) \approx 0.399$.

- Binomial distribution with $n = 50$ and p from question 14: **1**
- $P(X \geq 3) = 1 - P(X \leq 2)$: **1**
- Calculator command written down: **1**
- 0.399: **1** (using $p = 0.046$ gives 0.406; accept with no deduction if the rounded value was carried forward consciously, but advise using the unrounded value)

Part 7

Question 16 (4p)

$P(\text{double}) = \frac{6}{36} = \frac{1}{6}$; $P(\text{sum } 7) = \frac{6}{36} = \frac{1}{6}$; these do not overlap (a double has an even sum).

g	3	0	-1
$P(G = g)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{4}{6}$

$E(G) = 3 \cdot \frac{1}{6} + 0 - 1 \cdot \frac{4}{6} = -\frac{1}{6} \approx -€0.17$. $E(G^2) = 9 \cdot \frac{1}{6} + 1 \cdot \frac{4}{6} = \frac{13}{6}$; $\text{Var}(G) = \frac{13}{6} - \frac{1}{36} = \frac{77}{36}$; $\sigma(G) = \frac{\sqrt{77}}{6} \approx 1.46$.

- Probability distribution of G (net gain, so 3, 0, -1): **1**
- $E(G) = -\frac{1}{6}$: **1**
- Variance formula applied: **1**
- $\sigma(G) \approx 1.46$: **1**

Part 8

Question 17 (5p)

$f_p'(x) = \dfrac{2x}{x^2+p}$. $f_p''(x) = \dfrac{2(x^2+p) - 2x \cdot 2x}{(x^2+p)^2} = \dfrac{2p - 2x^2}{(x^2+p)^2}$. $f_p''(x) = 0 \iff x^2 = p \iff x = \pm\sqrt{p}$, and f_p'' changes sign there. $f_p(\pm\sqrt{p}) = \ln(p+p) = \ln(2p)$, and $\ln(2x^2) = \ln(2p)$ for $x = \pm\sqrt{p}$. So both points of inflection lie on $y = \ln(2x^2)$.

- f_p' correct (chain rule): **1**
- f_p'' correct (quotient rule): **1**
- $x = \pm\sqrt{p}$: **1**
- y-coordinate $\ln(2p)$: **1**
- Conclusion by substituting $x^2 = p$ in $\ln(2x^2)$ (or by eliminating p): **1**

Question 18 (4p)

$k_a(1) = 1-a$, $k_a'(x) = 3x^2 - a$, $k_a'(1) = 3-a$. Tangent: $y = (1-a) + (3-a)(x-1) = (3-a)x - 2$. For $x = 0$: $y = -2$ for every a. So all tangent lines pass through $(0, -2)$.

- $k_a(1)$ and $k_a'(1)$: **1**
- Equation of the tangent: **1**
- Simplified to $y = (3-a)x - 2$: **1**
- Conclusion: the point $(0, -2)$, independent of a: **1**

Question 19 (4p)

Gegeven: $ABCD$ cyclic; $S = AB \cap DC$ with B between A and S and C between D and S. *Te bewijzen:* $\triangle SBC \sim \triangle SDA$ and $SB \cdot SA = SC \cdot SD$.

1. $\angle SBC = 180° - \angle ABC$ (angles on the straight line ASB).
2. $\angle ABC + \angle ADC = 180°$ (opposite angles of a cyclic quadrilateral).
3. So $\angle SBC = \angle ADC = \angle ADS$ (as C lies on segment DS).
4. $\angle BSC = \angle DSA$ (same angle at S).
5. Two equal angles: $\triangle SBC \sim \triangle SDA$ (hh), with $S \leftrightarrow S$, $B \leftrightarrow D$, $C \leftrightarrow A$.
6. Corresponding sides: $\dfrac{SB}{SD} = \dfrac{SC}{SA}$, so $SB \cdot SA = SC \cdot SD$.

- $\angle SBC = 180° - \angle ABC$ with reason: **1**
- Cyclic quadrilateral theorem used correctly with reason: **1**
- Common angle at S and conclusion of similarity (correct correspondence): **1**
- Ratio of corresponding sides leading to $SB \cdot SA = SC \cdot SD$: **1**

(This is the *power of a point* theorem, E3 notes §5.)

Score table

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	Total
Max Score	4	4	5	5	3	4	4	4	4	4	3	3	4	3	4	4	5	4	4	75

Domain	Questions	Points
Functions and calculus	1, 2, 3, 6, 17, 18	26
Trigonometry	4, 5	8
Analytic geometry and vectors	7, 8	8
Parametric curves	9, 10, 11	11
Differential equations and algorithms	12, 13	7
Probability	14, 15, 16	11
Geometric proof	19	4